



Science, Technology and Modernity
Course Handout (Part II)
Semester II 2025-26

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : BITS F214

Course Title : Science, Technology and Modernity

Instructor-in-charge : Prof. Natasha Hazarika

1. Scope and Objective of the Course: This course is designed to provide students with a comprehensive understanding of how science and technology interact with societal structures, economic systems, and global governance. The course begins with an introduction to foundational perspectives on technological change and scientific knowledge systems, offering a basic orientation to how technologies are developed and adopted. It then moves to the examination of Intellectual Property Rights and Innovation, focusing on real-world issues such as patent disputes, corporate competition, and the strategic control of innovation. The third and final module explores Science, Technology, and Social Stratification, encouraging students to consider how power, inequality, and institutional interests influence technological priorities and outcomes. Overall, it critically assesses the societal responsibilities and consequences of scientific and technological work in contemporary settings.

2. Text Book:

Sismondo, Sergio, *An Introduction to Science and Technology Studies*, UK: Wiley-Blackwell, 2010.

Jacob, Margaret. 1997. *Scientific Culture and the Making of the Industrial West*. New York: Oxford University Press.

Friedman, Thomas L. 1999. *The Lexus and the Olive Tree*. New York: Anchor Books.

3. Reference Material- Books/Papers:

Book chapters, Journal papers, and newspaper reports will be shared as reading materials with the students.

4. Course Format: The course format will be interactive with regular class discussions and several group exercises and presentations. Students are expected to prepare in advance with the help of reading material provided and through self-initiative.





5. Evaluation/Grading:

Component	Duration	Weightage %	Date	Remarks
Class discussions (Linked to attendance)	In Class	5%	Everyday	
Mid Sem Exam	90 minutes	30%	06/10 AN1	CB
Assignments/ Internal Tests	To be Announced	30%	To be announced	OB/ CB
Compre. Exam.	180 minutes	35%	07/12 FN	CB

6. Course Plan:

Modules	Lecture Sessions	Ref. material	Learning Outcome
1. Fundamentals of technology and scientific knowledge systems	The Kuhnian revolution; Industrial Revolution; The Social Construction of Scientific and Technical Realities; Actor-Network Theory	Relevant readings	Critically analyze how scientific paradigms shift, how knowledge is socially embedded.
2. Intellectual Property Rights and Innovation	Knowledge protection; Innovation process; Spatial dimensions of innovation; Technology diffusion; Policy debates in innovation	Relevant readings	Understand how legal and institutional mechanisms structure technological innovation
3. Science, Technology, and Social Stratification	Stratification and Discrimination; Political economy of technology transfer; Role of power; Just Transition	Relevant readings	Understand how power, inequality, and institutional interests influence technological outcomes

7. **Chamber Consultation Hour:** 5-6 pm (Office: 6168L-by prior appointment).
8. **Notices:** Notices concerning the course will be displayed on Google Classroom.
9. **Plagiarism:** Plagiarism and cheating of any kind will be strictly punished through reduction of marks.

Prof. Natasha Hazarika
Instructor-in-charge
BITS F214

